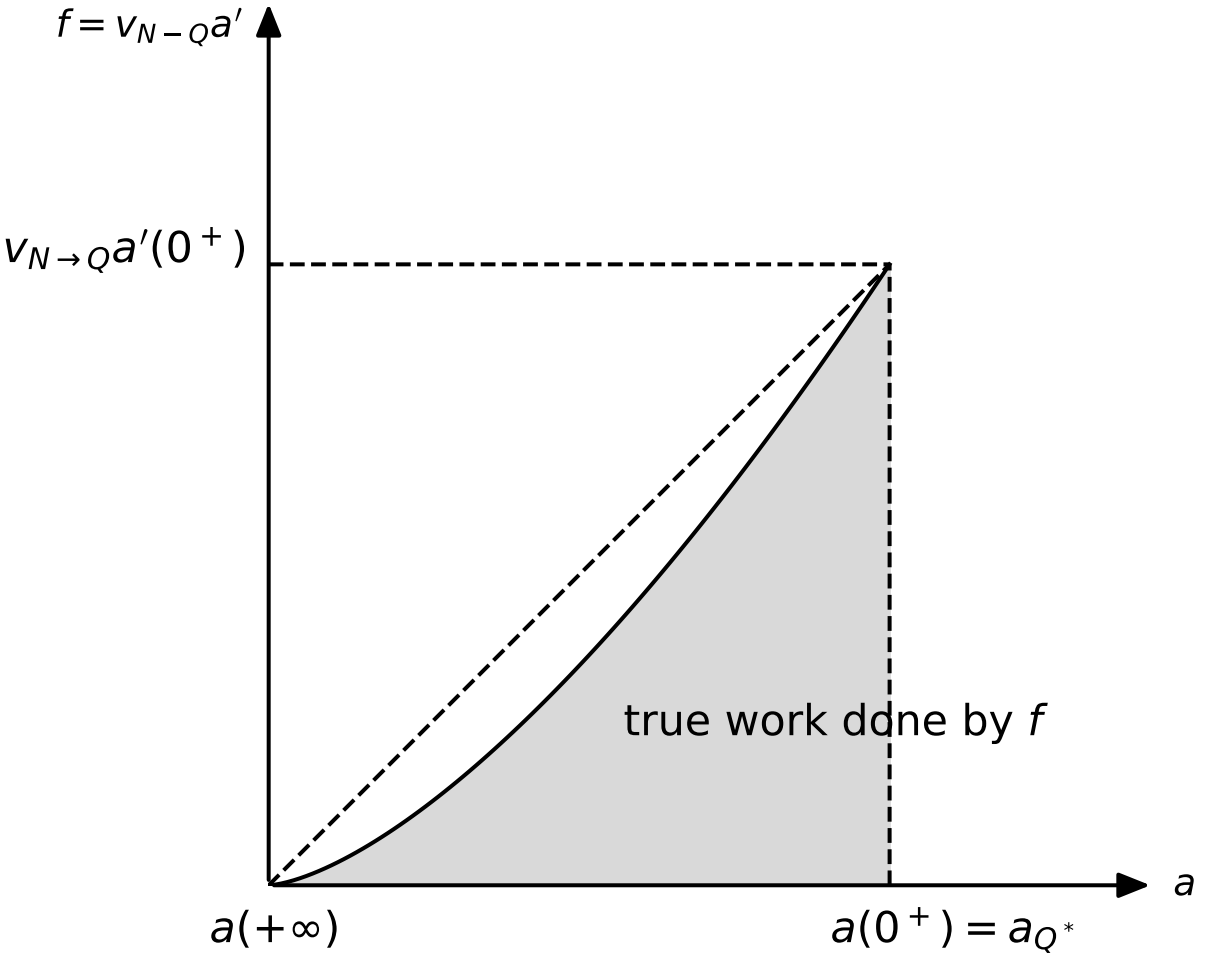
$f = v_{N-Q}a'$
$v_{N\to Q}a'(0^+)$
true work done by f
a
$a(+\infty)$
$a(0^+) = a_{Q^*}$